SMART MIRROR

PROJECT REPORT

submitted by

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In

Computer Science and Engineering



Department of Computer Science and Engineering

ST THOMAS INSTITUTE FOR SCIENCE AND TECHNOLOGY

KAZHAKUTTOM, TRIVANDRUM, KERALA

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DECLARATION

I undersigned hereby declare that the project report "**Smart Mirror**", submitted for partial fulfilment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of **Mrs. Divya Susan Mathew**. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the report entitled **'Smart Mirror'** submitted by '**Pratheesh VP'** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by him under our guidance and supervision. This report in any form has not been submitted to another University or Institute for any purpose.

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PRATHEESH VP

ABSTRACT

People greatly value their appearance, spending approximately an hour a day in front of the mirror during their morning and night routines. This is a significant amount of time where important things are taking place, but the mind is not working. It would be extremely useful to spend that time on the phone or computer completing any of the tasks mentioned above, but unfortunately it is difficult to do so while preparing for the day. A product is needed that can allow a person to efficiently complete everything they need to do to prepare for the day, all in one place and at the same time. Major problem of any existing mirror is displaying just the object in front of it or just the human face without having to interact with them. This project is developed with the intention that people spend quality time in front of the mirror. The goal of the Smart mirror is to provide a single easy to access location for a person to receive all the information that could affect how they prepare for the day. The smart mirror is an innovative technology that integrates the functionalities of a traditional mirror with advanced computing capabilities and internet connectivity. This project report aims to explore the design, development, and implementation of a smart mirror prototype, focusing on its hardware and software components, as well as its potential applications and future prospects. Moreover, the project explores the potential applications of the smart mirror beyond its traditional use as a reflective surface. It investigates the integration of voice recognition and gesture control, allowing users to interact with the mirror hands-free. The smart mirror can be employed in various settings, including homes, hotels, retail stores, and healthcare facilities, to provide personalized information, enhance customer experiences, and facilitate health monitoring. Throughout the project, attention is given to the challenges encountered during the development process, such as optimizing the mirror's visibility under different lighting conditions, ensuring data privacy and security, and addressing power consumption issues. The report presents the methodologies employed to overcome these challenges and suggests potential improvements for future iterations of the smart mirror. In conclusion, this project report presents a comprehensive exploration of the design, development, and implementation of a smart mirror prototype. It highlights the integration of hardware and software components, explores potential applications in different sectors, addresses technical challenges, and provides insights for future improvements. The smart mirror represents an exciting and promising technology that has the potential to revolutionize our daily interactions with reflective surfaces and provide personalized information and experiences.

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CHAPTER 1
INTRODUCTION

A Smart Mirror is meant as a smart home technology to allow convenient access to media and information that people might be interested in on a daily basis. The mirror will display customized information to the user to allow users to get some basic daily information at a glance. This will save users time in their busy morning routine which will, as a result, relieve stress and help people get out the door on time in the morning. The mirror will display information users may access in the morning via their phone or other technology in a way that is readily available as soon as they walk into the bathroom. This will allow the user to plan for their day while simultaneously completing their daily routine in front of the mirror.

While the user is preparing themselves for the day, they will be able to glance at their mirror and instantly retrieve important bits of information such as the current time, date, and weather conditions. These simple bits of information are commonly sought after in the morning while getting ready for the day or at night before going to bed.

Our Smart Mirror ensures that you remain at the forefront of the emerging smart-home revolution. You will take advantage of every opportunity to look at yourself in the mirror in order to be filled with weather information and today's date. With a brief look at the mirror, users will be able to check the current time, peek at the weather forecast, use the google assistant for interaction, be updated about the current happenings of the world and also able to see the notifications from the mobile. While the user is preparing themselves for the day, they will be able to talk to their smart mirror.

The smart mirror is an emerging technology that combines the functionality of a traditional mirror with advanced computing capabilities and connectivity to create an interactive and intuitive user experience. It represents a fusion of technology and design, transforming an ordinary mirror into a multifunctional device that provides real-time information, personalized content, and seamless integration with other smart devices.

The concept of a smart mirror stems from the desire to make our everyday routines more efficient and engaging. By incorporating a transparent display and a reflective surface, the smart mirror allows users to simultaneously see their reflection and interact with digital content overlaid on the mirror's surface. This integration of the physical and digital worlds enables a range of applications and possibilities. Ws\2ED2ED3One of the primary advantages of a smart mirror is its ability to provide

relevant information at a glance. Users can receive updates on the weather, news, calendar events, and social media notifications, all conveniently displayed on the mirror's surface.

The potential applications of smart mirrors extend beyond personal use. In retail environments, smart mirrors can enhance the shopping experience by providing virtual try-on capabilities, suggesting complementary products, and delivering targeted advertisements. In healthcare settings, smart mirrors can serve as valuable tools for health monitoring, displaying vital signs, medication reminders, and wellness tips.

The development of a smart mirror involves the integration of various hardware and software components. The mirror display utilizes special two-way mirror film that allows information to be displayed while maintaining reflectivity. A compact computer module, such as a Raspberry Pi, serves as the processing unit to run the mirror's software and connect to the internet.

The software aspect of a smart mirror involves creating a user-friendly interface and developing applications that provide the desired functionalities. Programming languages such as Python and frameworks like Electron are commonly used to develop the software stack for the mirror. The software not only manages the display of information but also enables customization, allowing users to choose their preferred widgets and personalize their mirror's interface.

Despite its numerous advantages and potential applications, the development of a smart mirror also poses certain challenges. Ensuring optimal visibility of the displayed content under different lighting conditions, addressing privacy and security concerns and optimizing power consumption are some of the key challenges that need to be overcome.

In conclusion, the smart mirror represents a fascinating and innovative technology that combines the utility of a mirror with the capabilities of a connected device. Its potential applications span various sectors, and its development requires the integration of hardware and software components. Overcoming technical challenges and addressing user needs will be crucial in unlocking the full potential of this transformative technology. The following sections of this project report will delve deeper into the design, development, and implementation of a smart mirror prototype, highlighting its hardware and software aspects, potential applications, and future prospects.

1.1 PROBLEM STATEMENT

The key companies are exponentially growing their dominance using the two main qualities efficiency and productivity to establish their products into the market. The fact that the product can

do multitasking and enhance productivity is the reason they are way ahead in the competition in the market. Their time management is an important factor in day-to-day production. This time management has been made possible only by integrating this technology into people's daily lives. The products such as PCs, Tablets, Smartphones and Smart Devices have given people the platform to use and provide access to the tools needed to be productive. The world around is changing daily. The various day to day activities that are being used through interactive computing via wirelessly connected devices is changing and improving the quality life standards. Based on this Interactive Computing and Wireless Technologies, many devices are now rising and with this Multimedia Intelligence, it is providing convenient, secure and comfortable, personal services and making maximum of users comfortable. There are Smart Cities, Smartphones, Smart Cars, and many more. The Smart Homes will operate through automation to open or close the windows based on commands provided to it. The vision of Artificial Intelligence has brought an interesting twist to the decade-old research initiatives in perceiving Smart Environments. The European Consortium proposed the Artificial Intelligence vision to promote a worldview where people are surrounded by smart interfaces offered by the interconnected diversified computing devices rooted in everyday tools. In this age of advanced technology, human life becomes easier and faster. This article describes the development and design of intelligent mirrors that will make us everyday life is easier and faster. Smart Mirror is a common mirror that has been improved using technology. The goal of Smart Mirror is to provide a simple information service such as news-feeds, weather, clock and so on and provide some basic AI elements such as real-time interaction with users, etc. The Smart Mirror CPU is the computer and the Raspberry Pi 3 framework that retrieves data via the Wi-Fi connection from the Web. In our rapidly developing world, you always have information at your fingertips - on your phone, on your phone, computer, maybe even on your watch. Keeping in touch with new information is for both care and daily life. With a variety of options, it is difficult to keep track of all your data flows. During the day, you may find yourself at a point where it is not practical or even impossible to remove the phone or computer and check the last update. You cannot engage in slower interaction. You need a display that you can access. However, aesthetics is as important as the display of information. Keeping an extra computer in your bathroom or in the corridor would be inconvenient and would not fit well into a modern space. An elegant and simple announcement that is easy for an average consumer is a necessity in today's world. The environment created can recognize and answer the actions and presence of beings. The Ambient Intelligence is thus seen as the driving force towards a high user friendly and user-empowered smart environment for providing effective help to human computer interactions.

1.2 INTERNET OF THINGS

The internet of things, or IoT, is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-computer interaction. A thing in the internet of things can be a person with a heart monitor implant, a farm animal with a biochip transponder, an automobile that has built-in sensors to alert the driver when tire pressure is low or any other natural or man-made object that can be assigned an Internet Protocol (IP) address and is able to transfer data over a network. Increasingly, organizations in a variety of industries are using IoT to operate more efficiently, better understand customers to deliver enhanced customer service, improve decision-making and increase the value of the business.

An IoT ecosystem consists of web-enabled smart devices that use embedded systems, such as processors, sensors and communication hardware, to collect, send and act on data they acquire from their environments [9]. IoT devices share the 3-sensor data they collect by connecting to an IoT gateway or other edge device where data is either sent to the cloud to be analysed or analysed locally. Sometimes, these devices communicate with other related devices and act on the information they get from one another. The devices do most of the work without human intervention, although people can interact with the devices for instance, to set them up, give them instructions or access the data.

IoT is revolutionizing the way we interact with the world, bridging the gap between the physical and digital realms. This technology holds immense potential to transform industries, improve efficiency, and enhance our daily lives.

At its core, IoT is driven by the concept of connecting everyday objects to the internet, enabling them to communicate and share data. These objects, often referred to as "smart devices," can include anything from smart thermostats and wearables to industrial machinery and smart city infrastructure. By collecting and analysing data from these devices, organizations and individuals can gain valuable insights, automate processes, and make data-driven decisions. One of the key benefits of IoT is its ability to improve efficiency and productivity. Connected devices can monitor and optimize various processes, such as energy consumption, inventory management, and predictive maintenance. For example, in manufacturing settings, IoT-enabled sensors can collect real-time data on machine performance, helping identify potential issues and preventing costly downtime.

IoT also has the potential to enhance safety and security. With connected devices, it becomes possible to remotely monitor and control security systems, access controls, and surveillance cameras. In smart

homes, IoT devices can detect smoke, gas leaks, or unauthorized access, triggering immediate alerts to homeowners and emergency services. Furthermore, IoT plays a crucial role in the development of smart cities. By integrating various systems and infrastructure, such as traffic management, waste management, and energy distribution, cities can become more sustainable, efficient, and liveable. IoT-enabled sensors and networks allow for real-time monitoring and optimization, leading to reduced congestion, improved resource management, and enhanced quality of life for citizens.

However, the widespread adoption of IoT also brings challenges that need to be addressed. Security and privacy are major concerns, as the vast amount of data collected by connected devices needs to be protected from unauthorized access and misuse. Additionally, interoperability and standardization are crucial to ensure seamless integration and communication between different devices and platforms. In conclusion, IoT is a transformative technology that connects the physical and digital worlds, enabling devices to collect and exchange data over the internet. Its potential for improving efficiency, enhancing safety, and transforming industries is vast. However, addressing security, privacy, and interoperability challenges is essential to fully unlock the benefits of IoT. As the technology continues to evolve, the integration of IoT into various aspects of our lives will undoubtedly shape the future and create new opportunities for innovation and growth.



Fig 1.1: Internet of Things

1.3 PLATFORMS

Recently, intelligent mirrors such as Magic Mirror and Home Mirror have been developed people in the Maker community with various levels of interactivity. So far, anyway the characteristics of these mirrors are limited. This latest project describes how a smart mirror works was built from scratch with a Raspberry Pi for hardware and custom software running Raspbian, a Linux distribution. The goal of the project was to create a Smart Mirror device People could interact with the technology, but also develop the technology so that you can make it happen Install and develop your applications for this. Time management is a very important factors for success and efficiency in a person's everyday life. With the increasing world of technology, maintaining an efficient schedule in our day-to-day life has become both easier and more difficult. Keeping updated with meetings, news, social media, and other tasks are made easier through technology such as PCs, tablets, and smartphones but they also provide distractions that mostly interrupt the user's routine. Technology has become one of the important objectives in the day that time must be allotted for. In the limited time of the day, technology needs to be designed in order to function within the schedules of the user and not be an additional element to it. Multitasking is the key to efficient time management involving technology. Anyone in the business or the academic world will agree on the point that each second in a day counts and hold utter importance. This elongates as well to the continuing trend of integrating touchscreens and internet-connectivity into everyday appliances such as washing machines and refrigerators. The idea of a smart home is the direction towards the smart world to which lots of corporations are heading and while the smart kitchen modules have been turning eyes in its direction, the smart bathroom has not. Alongside the kitchen, the bathroom is one of the busiest rooms in any home, so it is a great place to expand the smart home initiative. It may be desirable to be able to interact with the mirror rather than simply being a static display of information. Most smart devices use touch functionality for interaction. However, since this is a mirror, other methods are needed to avoid fingerprints in the mirror. The graphic display offers a limited graphic area. As a result, the functionality it can provide is limited because only so many modules can be represented in this display. By offering interaction methods, such as voice commands and gestures, the user can start the mirror to view information on demand. Instant access and updated information are driving the current generation. Forget bringing gadgets like smartphones, tablets, etc. in the bathroom and risking damage. The smart mirror will show the information with the command of your voice or control from the hand. The smart mirror is the outcome of our team discussion on how to overcome all these issues and develop something that is operational as well as a working model.

1.4 HARDWARE

The smart mirror should offer the benefits of using modern and advanced technology while integrating smoothly into the basic bathroom routines of people. The smart mirror must be simple and instinctual as much as it can be. The smart mirror is basically implemented to combine technology and the need for information into the user's every day plan. With the mirror in position, the user can interact and obtain the information they want during their normal bathroom routines. This smart mirror aims to reduce and eliminate time and effort a particular user spends in their daily routine to check their PC, tablet, or smartphone for the information they require. The mirror will prepare relevant information with minimal effort from the user with the goal of luxury that they must maintain. The mirror would be an improvement over the current normal utilization of mirrors in most modern bathrooms. The mirror can do the functionality and thinking for the user. First, it will turn On/Off by the commands given to it. Then, it will update with the user's calendar schedule knowing its location, to-do lists, news, and weather forecast. The information would not be displayed in front of the user's face, but quietly displayed on the sides of the mirror to still allow the actual use of the mirror. The mirror provides the most common information that people usually check on their smartphones or tablets, such as weather, news, and schedules. This allows the users to read, think, and plan their day while getting ready in the morning. The smart mirror requirements and specifications are inspired from people's everyday devices that they use like computers, tablets, and smartphones. The mirror integrated similar features from each to give the user what they would expect out of a modern smart device The smart mirror was run through a central computer that took in the data from the various applications and peripherals which in turn were used to provide the user with its key functionality. The mirror uses a microphone for taking input. The RPi is used as a built-in computing device for the smart mirror. The Smart Mirror user interface has a number of standard applications important information for the user, including weather, news, Calendar, music and clock. Depending on the user's request, you can also incorporate additional components. The smart mirror has developed voice control through a voice recognition system in the user interface. The project can be improved by adding interactive 6 touchscreens, Geo-Location, Alexa and many other features. The Smart Mirror was developed in 4 months, starting with the software and finally integrating it with the hardware. The results were good because a higher level of interactivity has been achieved by being able to use voice commands, gestures and mood detection. Many problems arose in the hardware and software side of the project, for example, the microphone couldn't recognize the commands and hence voice recognition being unreliable. This project also includes the construction of an intelligent mirror. The smart part is provided by the software. The physical device built would rather be an intelligent mirror, depending on the distinction made between the intelligent and intelligent mirrors made during

the introduction. There are many features to consider when creating the mirror: weight, mirror size, screen size, hardware selection, power supply, mounting technique, heat generation and aesthetics. A mirror is also a decorative piece that is functional and aesthetically pleasing is considered important. The smart mirror is certainly not a true consumer product. There are very few of them manufactured and ready to sell smart mirrors on the market. The ones that are there are a lot different in terms of functionality, development and price. It will certainly take a while the big houses of smart homes to come behind this product and make it mainstream Customers. Each product had a common feature, namely health management, e.g., as a weight. This is something that our intelligent mirror does not have a direct focus on and perhaps If our project becomes public, design would be something to change. This report presents the Implementation of Smart Mirror Using Raspberry Pi and Artificial Intelligence. A Smart Mirror displays Time, Date, Weather, Notification and News. The benefits of using a Smart Mirror are that it makes life easier as: Need to look at phones, every time need to check Time, Weather is also reduced. The Smart Mirror will help in developing the initiative of Smart House with Embedded Artificial Intelligence finding application in industries. The Interactive Mirror is being developed such as the Mirror with properly embedded Artificial Intelligence for providing features such as Weather, latest news updates and their headlines and time according to the location of the mirror.

CHAPTER 2
LITERATURE SURVEY

2.1 INTERACTIVE MULTIMEDIA MIRROR SYSTEM DESIGN (Jun-Ren Ding, Chien-Lin Huang, Jin-Kun Lin, Jar-Ferr Yang, Fellow, IEEE and Chung-Hsien Wu, Senior Member, IEEE)

At present, human beings need to be connected and access information easily. Whether through television or the Internet, there is a need to be informed and in touch with current issues that happen around the world. In such a way that the human being needs comfort in his life and has adapted the technology based on their needs [1]. In recent years, intelligent environments are a popular trend in the concept of Internet of Things (IoT), within which several analytical methods of data such as machine learning and sensor technology have regained importance in academic research whose objective is to provide services of the type not input values, for example, in a domestic environment can be install a mirror with intelligent capabilities much like smartphones. For85 the importance of the internet of things, lies in the need to have control over almost any object that can be connected to the internet, where the user can manage, observe, integrate and interact it with other object to obtain data by means of special sensors and that can facilitate making decisions by the user according to the domain where the solution is implemented. For Kaori the Smart Mirror is a screen that has a typical functionality of a mirror and also displays multimedia data through a glass screen which can be integrated perfectly in a home [3]. For Derreck, the Smart mirror is a device that works like a mirror with additional capacity to display multimedia data such as text, images and videos. Based on the definitions found, we can define a Smart Mirror as a device built with features of a normal mirror with intelligent-type functionalities whose purpose is to provide interactive information services that will help you start your day [4]. According to Chidambaram, efforts have been made to add technologies in the Smart Mirror to allow multiple tasks simultaneously in order to save time. They develop a mirror with services based on the biometric identification of the user by extracting facial features for recognition, by using a facial recognition algorithm and with images housed in the database created by them, which serves as data of training for the algorithm [5]. For Cicerone, there are several research groups that are working on the search for alternatives to implement home automation systems offering better user interfaces. They propose the development of an innovative device that incorporates interactive leisure and information services offered in the interface of a mirror. They suggest a personalization methodology through the creation of a recommendation algorithm, which associates instances of the user's tastes with a service, for example, in the case of

television a personalized programming guide, personal motivator or activity reminder [6]. For the prediction of user tastes, classic and modern filtering techniques such as collaborative filtering and content-based filtering can be used. [7]. In this way, we propose as a solution the implementation of a smart mirror in order to help optimize people's time before starting the activities of their day. Providing relevant information that allows them to be connected with the current world. Likewise, it is proposed to add to this solution an authentication system for facial recognition, a recommendation system for the classification of news according to the interests of the user and with voice interactions that allow offering a personalized experience.

2.2 SMART MIRROR: A LEAP IN HOME AUTOMATION (Mr. P. Mathivanan (Assistant Professor) Department of Information Technology ManakulaVinayagar Institute of Technology Puducherry, India, Anbarasan G (B. Tech) Department of Information Technology ManakulaVinayagar Institute of Technology Puducherry, India)

The story of “Snow White” features a magic mirror that knows everything and provides updated information to the user, i.e., the Queen, in an interactive manner. In this paper, the design and implementation of an interactive multimedia mirror system, called the magic mirror, are described. The magic mirror figured with the techniques of speech recognition and synthesis, 3-dimensional (3D) graphic generation, Internet connectivity, and multimedia services is integrated. With regard to the hardware system, the magic mirror equips with a microphone and a pair of loud speakers for speech interaction and audio services. For human visual interaction and graphical presentations, it also includes a liquid crystal display (LCD) display and CCD camera, which are covered by a plate of reflective glass. As shown in Fig. 1(a), the reflective glass has the following characteristics [8]. If the luminance in front of the glass is greater than that behind it, the lighter side exhibits the attributes of a mirror, while the darker side is transparent. Once the reflective glass is installed in front of the LCD panel, the magic mirror, acts like a real mirror if the LCD display is switched off. On the contrary, the magic mirror displays the image directly from the personal computer, if the LCD display is switched on. achieve the reflective mirror, the glass coating with heat insulation polyethylene can also produce a similar effect. The CCD camera and infrared sensors are installed on top of the LCD display and covered by the reflective mirror. As shown in Fig. 2, the hidden CCD camera can precisely capture the image without the affection of a reflective mirror. The microphone could be installed in any other appropriate location for better speech reception. For remote command, a personal digital assistant (PDA) with touch panel control or a Bluetooth microphone for speech commands is also considered. The magic mirror, which can be a smart living product, should provide as many services as possible. However, several practical factors, including hardware cost, operation

scenario, computational complexity, environment and personal adaptation should be taken into account. Clearly, nobody wants to purchase a magic mirror that can only deliver a single service. The magic mirror should be able to retrieve Internet information and provide real-time personal services with seamless control interaction.

2.3 IMPLEMENTATION AND CUSTOMIZATION OF A SMART MIRROR THROUGH FACIAL RECOGNITION AUTHENTICATION (Ivette Cristina Araujo García Computer Science Universidad Peruana de Ciencias Aplicadas Lima, Perú, Alfredo Barrientos Padilla Computer Science Universidad Peruana de Ciencias Aplicadas Lima, Perú)

Smart appliance design figured with multimedia intelligence to provide comfortable, convenient, and secure personal services in home becomes increasingly important in the era of information communication technology. Based on pervasive and ubiquitous computing, several research works have successfully integrated human computer interface technologies to realize several advanced smart living products [1]–[7]. In [2], [3], the monitors equipped with a mirror system have been developed for smart houses. [2] utilized a projector and a charge coupled device (CCD) camera to implement a magic mirror, the proposed system still has three major limitations. First, it is infeasible to implement such applications in a normal house owing to the large space requirement. Secondly, an overlapped image is formed between the project and mirror. In [3], an aware mirror was developed for application in a bathroom; its features include user detection, display method, and content function. Acrylic plaster was utilized instead of glass for the mirror. However, acrylic plaster degrades easily by scratch and heating. Moreover, placing the radio frequency identification (RFID) device in a toothbrush is a rather excessive design since toothbrushes are typically not carried personally. Conversely, if all the members in a household put all their toothbrushes together, the personal information service will not be properly activated. Philips Home Lab is a test bed for the creation and innovation of technologies in a domestic environment. Among its projects is the creation of an interactive mirror in the bathroom that provides personalized services according to the user's preferences. For example: children can watch their favourite cartoons while brushing their teeth, provide TV channels, check traffic information, etc. The authors suggest that the device can be constructed from an LCD screen combined with a reflective surface and connected to a central processor that provides the aforementioned services [8]. The company The Fung Group, conducted an investigation on smart mirrors that focuses on the retail sector or clothing marketing. Being one of the highlights the Memomi's Memory Mirror, intelligent mirror whose functionality is similar to a virtual clothing tester, it allows customers to change sizes, colours and styles of outfits in real time and, of course, share them through mail or social networks It has a camera and sensors at the top with

which it follows the movements in real time can record videos that offer 360-degree images when the person is in front of the mirror trying clothes through it [9]. Michael Teeuw developed an open-source intelligent mirror platform called MagicMirror², he is considered as the pioneer in developing a device with a low-cost Raspberry Pi board, the final product of this project is a framework that has different modules, among they identify the functions of calendar, weather, news, alerts, as well as facial recognition. The smart mirror we build in our project implements all of the functions mentioned above. In addition, it has features for news recommendation and interaction by voice commands, these two features seek to provide a more sophisticated user experience and allow a natural interaction between the mirror and the user. [10].

The process of face recognition begins with the capture of the image by means of a camera, this image needs to be pre-processed to improve the contrast. Generally, an image is a collection of pixels, whose value is given according to the intensity that is analysed from the upper left pixel to the lower right pixel. The intensity value is distributed over the entire image using histogram equalization technique. [5]. For Doychin, referral systems have become an essential part of our day-to-day life, especially when it comes to handling large amounts of information available online. The use of these systems improves the user experience either when making a purchase like Amazon, eBay or reviewing news such as the case of Google News and BBC [11]. For García, recommendation systems have proven to be an important way for people to discover information, products and services that are relevant to their needs. These are complemented by search services based on more conventional queries, offering more proactive information discovery, often based on a profile of short or long-term user preferences. These systems can now be found in many applications to provide users with a list of recommended items they would like. For example, popular online retailer Amazon offers users a list of product recommendations in different categories, while Last.fm does it for music and Netflix for movies [12]. The recommendation systems of news articles to the readers, has a series of unique characteristics compared to the recommendation systems of films or products where the profile of the user is constructed from the action of rating the article or the registration of the Preference history that is obtained through interaction on the page. In many traditional application domains of recommendation systems, a user profile is constructed from a user transaction or preference history, for example, movies that have been rated or products purchased. The profile is associated with the user as they interact with the site, and feeds into the creation of personalized recommendations. In the domain of the news, it is not common to have detailed profiles of the user, since users are often not required to log in or create programs [13].

2.4 SHOE CUSTOMERS' BEHAVIOUR WITH NEW TECHNOLOGIES: THE MAGIC MIRROR CASE (Claudia Redaelli 1, Raffaella Pellegrini 2 , Stefano Mottura 1, Marco Sacco 1 1 ITIA-CNR Institute of Industrial Technologies and Automation Via Bassini 15, 20133 Milano, Italy, claudia.redaelli, stefano.mottura 2 Universita Cattolica del Sacra Cuore Milano LargoA. Gemelli 1,20123 Milano, Italy)

European shoe manufacturing sector suffers a high pressure due to the eastern countries competition. Maintaining competitiveness can appear a chess game in which western countries have to strategically find the appropriate solutions. Dorothy is a research project funded by the European Commission within the ih Framework Programme under the NMP theme. It promotes a response based on the transformation of the shoe industry and related business model meant to strengthen Europe's ability to compete in terms of high added value for the customer. This foreseen transformation deals with the development of tools for the design of customer driven adding value shoes and with the realization of tools for the design and reconfiguration of flexible multi-site and multinational production plants. Since the customer is the centre of the innovation process based on new technological approaches, this paper analyses the customer approach to the customized product and in particular the impact of Augmented Reality (AR). The Magic Mirror uses the AR (mixed of real and virtual images) for allowing the user to try-on a pair of virtual customized shoes. This new trial step implies a new purchasing path and a different approach with the product. This paper presents the Magic Mirror case study and research made with a new questionnaire called "Flow for Presence Questionnaire" measuring the state of the user during the trial.

Internet services transformed our lives by connecting us more without difficulty to share information and other people in the essential world. An aware mirror was developed for application in a bathroom; its features include user detection, display method, and content function. Acrylic plaster was utilized instead of glass for the mirror. However, acrylic plaster degrades easily by scratch and heating. Moreover, placing the radio frequency identification (RFID) device in a toothbrush is a rather excessive design since toothbrushes are typically not carried personally. Conversely, if all the members in a household put all their toothbrushes together, the personal information service will not be properly activated. Philips Home Lab is a test bed for the creation and innovation of technologies in a domestic environment. Among its projects is the creation of an interactive mirror in the bathroom that provides personalized services according to the user's preferences. For example: children can watch their favourite cartoons while brushing their teeth, provide TV channels, check traffic information, etc. The authors suggest that the device can be constructed from an LCD screen combined with a reflective surface and connected to a central processor that provides the

aforementioned services Nowadays everyone should use a Smartphone and then this process has appeared and morphed into the Internet of Things concepts.

2.5 HOME AUTOMATION USING SMART MIRROR (Si Liu, Luoqi Liu, Shuicheng Yan Department of Electrical and Computer Engineering National University of Singapore)

The proposed system is exploring to create new technologies that are used in new different ways. But nowadays many different kinds of innovation are currently providing less interaction and also provide more useful information. So we have researched and designed the device for home automation using smart mirror. Relevant items are displayed by wall mounted mirrors in which user analysis to weather report, scheduling and other fields of home appliances should be used to control the device. It should solve the problems that many people experience every day by using wall mounted mirror. The user will may know the next morning whether it will show, it can help to plan their easily travel day to day life. The field of mirror technology has witnessed the beginning of revolution taking shape in world. The greatest innovations in technology are the smart mobile phone and IOT sensors. The most one of the features and characterized are sophisticated operating system allow users to peruse the internet for listen watch and record video using GPRS.

All the content in mobile phone such as personal photos, contact details, call histories and private messages used to store the in build large size of devices. To express development to a many users can shift in Smartphone device user replace of personal computers. The Smartphone usage is compared to personal devices is very efficient and comfortable for many users. In statistical analysis marketplace explore has surpassed the sales of laptops compared to mobile phones. International data corporation analysis the statistical of the mobile phone user details and shipment of mobile phone manufactured at various companies. In 2015 shipments in the second quarter of 337.2 million smart phone worldwide reached and compared to 2014 shipments 11.6 % increased by same quarter. The incredible increase in the number of consumers for buying smartphone devices. In 2015 the second quarter is the highest quarterly on record in mobile phone shipped predicated to increase to 1928.4 million increased in 2019. This paper focuses on the new method of user interaction with the smart mirror technology. This smart mirror technology is powered by firebase cloud services. The web application and mobile application has been developed by most popular platform firebase. The firebase joins together with various other Google services to present broader goods and range for developers. The following service Firebase consists of the services such as, Real-time database, and cloud hosted database comes under the firebase platform.

2.6 AI SMART MIRROR (Abdullahil Kafi, M. Shaikh Ashikul Alam and Sayeed Bin Hossain)

This project was proposed by students of BRAC University Dhaka for improving the day-to-day life of the people by saving their time and providing them quick information and basic updates. The updates and information were related to the local, national or international news, weather and calendar. The specifications and requirements of the Smart Mirror have been inspired by people's everyday devices, including tablets, PCs and smartphones. The integrated features were similar to what users expect from a modern and smart device. Smart Mirror includes some basic features like clock, calendar, newsfeed, etc., essential for everyday life, making it even easier for users to access these features. There are many previous projects related to Smart Mirror, but few where artificial intelligence is integrated. Users can modify the mirror in real time via artificial intelligence and access information available on the Internet, such as: on maps, weather, newsfeeds, etc. The CPU used the Raspberry Pi 3B minicomputer and all the software aspects were installed in the OS. The CPU captures video and audio data from the camera or microphone and through the face. The recognition and the voice recognition model identify users. As soon as the Smart Mirror detects the presence of a person, it is "woken up" and displays the basic functions such as time, calendar, weather update and so on. Smart Mirror also has some basic AI features. Users can interact with the mirror in real time and search for information on the Internet, ask questions, perform specific tasks, such as: B. set alarm clock, reminder, etc. Using the smart mirror. All information is displayed on the LCD screen to which it is connected the raspberry pi. After configuring all the hardware and software components, the model was checked for the result. The first start was unsuccessful because artificial intelligence did not respond to a voice command due to the ffmpeg error and the voice recognition package. However, the library and package were reinstalled, so the model was restarted. This time he might recognize the voice, but the model responded very slowly. Subsequently, the problem with booting the system was that the AI included the different types of entities, as the app was trained with wit.ai. However, during the overclocking of the Pi processor, a good result was achieved. The GPU has been lowered to get the maximum speed of the Raspberry Pi. Efforts have been made to develop an intelligent and efficient mirror that optimizes our working hours and increases our daily productivity. Smart Mirror will play an important role in IoT and Home Automation. Not only can it function like a normal mirror, but it also provides other features like weather forecasts, calendar, time of day, etc., which makes it more desirable. The mirror's functionality can be expanded by connecting it to other home appliances, mobile devices, etc. Smart Mirror is a great example of how artificial intelligence can be integrated into appliances to make our life easier, more efficient and more enjoyable. In the future, Smart Mirror can be made smarter by updating artificial intelligence.

2.7 FITNESS MIRROR (Daniel Besserer, Johannes Bäurle, Alexander Nikic Frank Honold, Felix Schüssel & Michael Weber)

This mirror is made up of a variety of sensors that read a 3D scan of the body and look for the anomaly of the formations. It also covers the area of the body that is prone to injury. It also recommends exercise plans to be fit. FitMirror detects the presence of a user with Balance Board and Kinect v2. When the Balance Board detects a user who steps on it, it sends a message to the main application to turn on the display and activate the Kinect camera. At this point, Kinect checks if the user is still in front of the mirror. If the user is not in front of the mirror for a while, Kinect detects it and sends a message to the main application to disable FitMirror. To distinguish which user is in front of the camera, the MAC address of the user's smartphone is loaded. To avoid further burdens for the user, the interaction with the mirror should be as simple and natural as possible. For this reason, the user has the possibility to use the tactile or voice commands to interact with the system. As an output mode for the image and sound of the mirror system, the display and the mirror speakers are used. When synchronizing with a smartphone with FitMirror, the FitMirror app loads the passages and caloric data burned from the GoogleFit database and sends them to the main application, where that data is stored in a database. Furthermore, the Balance Board measures the user's weight and sends the measured values to the main application, where they are stored in the same database as the other data. The stored data can be viewed as diagrams. For this function, the FitMirror main application loads the data from the last seven days and displays them. The data displayed is the weight, steps and calories burned by the user. FitMirror offers ten different exercises that the user can perform in front of the mirror. These ten exercises are divided into two types: fun exercises and normal exercises. Fun exercises are designed to increase user satisfaction and normal exercises increase the motivation and fitness of the user. With the help of the Kinect camera and the FitMirror Balance Board it is able to recognize the different exercises. The Kinect camera analyzes the user's skeleton and may differ between the various gestures implemented. The Balance Board measures changes in the user's weight distribution to determine specific movement patterns and determine exercises. The performance of each user exercise is stored in a list of top scores. Users can view all the executions of the exercises in this list. In addition, the user can view the services of other registered users of FitMirror to compare them with their services. Another feature of FitMirror is that users can challenge other users with an exercise. After a user has completed an exercise, he can challenge another user with the score obtained by selecting a user from the user list. When the challenged user uses FitMirror, he receives notification of a challenge request. You can accept or reject the request

2.8 SMART MIRROR (Microsoft)

This mirror was proposed by Microsoft in 2016. This smart Mirror runs on Windows 10 IoT Core on Raspberry Pi 3. This is supported by Windows Hello cognitive services. This it was an open-source project. The web app has been opened GitHub repository so that everyone can create their own smart Mirror. The mirror shows traffic information, weather and supports voice recognition.

2.9 RAPHAEL DYMEK (Apple)

This Mirror prototype by Apple is based on iOS 10, which mirrored the display of the iPhone. The mirror can start all mobile apps desired by the user. This mirror sleeps 12 every 45 seconds the ideal situation This is a smart touchscreen mirror.

2.10 MAGIC MIRROR FOR TEACHING (Stuckler)

There had been several projects which help in making the teaching and understanding easy with the use of technology. In this project, they proposed an idea of introducing a smart mirror which could help the educators in teaching various subjects, like anatomy with easy and practical implementations of its procedures. They used a color display device Camera, depth camera and Microsoft Kinect v1 from the Xbox 360 game console. When the user is in front of the display, the mirror loads computerized tomographic displays or 3D models of organs to give the user the illusion of looking into his body. With simple hand gestures, the user can modify the sections or zoom in and out. Other magic mirrors use single-board information technology, e.g., the Raspberry Pi and check with the spy film. In most cases, these mirrors show nothing but information

2.11 INTERACTIVE MIRROR (Santa Clara University)

The motive of this project was to develop a discrete display platform on a mirror. Their requirements were quite different from many of the existing projects. Initially, they wanted to display all the information on the screen like the weather, date-time, i.e., calendar, news updates and some other information which are required to the user. They wanted their system to be controlled by something which doesn’t require direct input, like they don’t want to enter some input and then wait for the results. Instead, they attached a small but high-quality camera which overcame this difficulty and gave proper results after getting the inputs. The most fundamental part was to connect the mirror to

the internet in order to receive the data. This was done with the assistance of the WiFi 16 module which provided the required amount of network strength for getting connected to the internet. They wanted their setup consisting few basic modules of artificial intelligence and wireless fidelity. In order to accomplish this problem, they used various APIs and saved the respective data in the database. They also wanted to control alternative input methods and was done by using the algorithm of face detection which helped them to focus on one particular input at a time and providing the respective results on the mirror screen. They also implemented auto-sleep functionality in the mirror in which the automation process in the mirror gets automatically turned off when no input is provided to it for some period, and automatically turns on whenever it gets any input.

2.12 INTERACTIVE MIRROR (Lit Studios)

A touch & gesture functional mirror created by two technology enthusiasts, Alpay Kasal and Sam Ewen. The user touches the mirror, which has an integrated touch screen to interact with it. Unlike our Smart Mirror project, just a point of touch is detected because it emulates a mouse. This mirror is less about data than artistic images. This mirror also had access to the wireless module in order to show the images and the drawings which were directly linked from the web. These pictures were searched with the help of the crawlers in order to provide the efficiency to the user. Users of the demonstration video show various types of 2D drawings and games displayed with a projector. The fact that you emulate one the mouse is beautiful, because the expandability and the range of functions are capable. However, this is still different from the complex Smart Mirror because it is not meant to solve anything, but it would help in increasing the quality of the display of the mirror.

CHAPTER 3
PROPOSED SYSTEM

Implementation of an interactive multimedia mirror system is called Smart Mirror. The Smart Mirror can be implemented in existing personal computers or hand- held devices with normal peripherals and regular reflective glass by integrating image/speech processing, internet connectivity, and 3D and multimedia. In this era of evolving technology, human life is becoming simpler and time efficient. This report depicts the design and development of smart mirrors which will make our everyday life easier and time efficient. Smart Mirror is a simple mirror which has been enhanced by the help of technology. The aim of the smart mirror is to provide an easy way to information services such as news feeds, weather, clock etc. It also provides some basic AI features like real time interaction with users and so on. The Smart Mirror CPU is the Raspberry Pi 3 computer and the framework that retrieves data from the web through the Wi-Fi connectivity. In today's society information is available to our phones, our laptops, our desktop and more. The one that concerns the common man is now it can be used to make day to day life easier and faster. This project which would collect real world machine data and data would be transmitted from the machine and managed by the Raspberry Pi Board. This Smart Mirror is the latest version of Raspberry Pi. The mirror provides most basic common features such as Weather of city, Updates of news and Headlines corresponding to Location. The user can read the news feed or check today's weather while dressing or shaving in front of the mirror. The aim of the project is to achieve such a device, and incorporate cutting edge features. The Pi will connect with the internet to retrieve real-time information to display on a monitor. This smart mirror aims to reduce and possibly eliminate the need for the user to make time in their daily morning or nightly routine to check their PC, tablet, or smartphone for the information they need. The mirror will provide the information with little to no effort from the user with the goal of not being a burden that he or she must maintain. The mirror wouldn’t be another activity, rather an enhancement to the already common use of mirrors in most modern bathrooms. The mirror will do the thinking for the user. Then, it will update with the user’s calendar schedule, to-do lists, Twitter, news, and weather. The information wouldn’t be thrown in the user’s face, but unobtrusively displayed on the edges of the mirror to still allow use of the actual mirror. This allows the users to read, think, and plan their day while getting ready in the morning or night.

3.1 MODULES

3.1.1 Hardware modules

The hardware modules consist of the following:

Hardware components are:

- Raspberry pi 3
- SD card - 16gb
- LCD panel display – 19 inches
- Power Supply 5V micro-USB
- Connecting wires
- One-way reflective mirror
- Mirror frame

3.1.2 Software modules

The raspberry pi used, runs as the heart of the system. The pi is connected with the internet so that it uses an API for fetching news from the website and it also fetches the temperature of the specified location. The fetched news is then displayed on the screen. It also shows the notifications of the user whose android is connected to the mirror. It also shows the important date remainders. The mirror also contains a personal assistant which is capable of fetching any details from the web other than the elements displayed on the screen. These elements are all added on to the UI in which user interacts leaving some space in display to act as a mirror so that user can use the mirror for grooming

3.2 SENSORS AND COMPONENTS

3.2.1 Raspberry Pi 3

The Raspberry Pi 3 is a single-board computer developed by the Raspberry Pi Foundation. It is the third iteration of the Raspberry Pi series and was released in February 2016. The device is small, affordable, and can be used for a variety of applications, from simple projects to complex applications. The Raspberry Pi 3 is powered by a 1.2GHz 64-bit quad-core ARM Cortex-A53 CPU, making it much faster than its predecessors.



Fig 3.1 Raspberry Pi 3

.It also has integrated wireless connectivity, including Wi-Fi and Bluetooth, which makes it easy to connect to the internet or other devices without the need for additional hardware. The board has a number of ports, including four USB 2.0 ports, an HDMI port, a 3.5mm audio jack, a microSD card slot, and a 40-pin GPIO header. This allows for easy connectivity to peripherals and expansion boards. The Raspberry Pi 3 also features 1GB of RAM, which is sufficient for most applications, as well as an improved graphics processor that can handle high-definition video playback. One of the key advantages of the Raspberry Pi 3 is its versatility. It can be used for a wide range of applications, including as a media centre, a web server, a gaming console, or as the brain for a robot. Its low power consumption and small size also make it an ideal choice for projects that require portability. The Raspberry Pi 3 runs on a variety of operating systems, including Raspbian, Ubuntu, and Windows 10 IoT Core. Raspbian is the official operating system for the Raspberry Pi and is based on the Debian Linux distribution. It is optimized for the Raspberry Pi's hardware and includes a range of software packages and tools to get started. The Raspberry Pi 3 also supports a range of programming languages, including Python, C++, and Java, making it easy to develop software for the device. This makes it an excellent choice for educational purposes, as it can be used to teach programming and electronics to students of all ages. In terms of performance, the Raspberry Pi 3 is more than capable of handling basic computing tasks. It can run web browsers, word processors, and other productivity software with ease. It can also handle more resource-intensive applications, such as video editing software, although performance may be limited by the amount of RAM available. The Raspberry Pi 3 has also spawned a thriving community of enthusiasts and developers, who have created a vast ecosystem of software, hardware, and projects. There are thousands of projects and tutorials available online, covering everything from basic programming to advanced robotics. Despite its cost and small size, the Raspberry Pi 3 is a powerful and capable device. Its versatility, ease of use, and low power consumption make it an excellent choice for hobbyists, educators, and anyone who wants to

experiment with computing and electronics. Whether you're a beginner or an experienced developer, the Raspberry Pi 3 is a device that can be used to learn, create, and explore.

3.2.2 Microphone



Fig 3.2 Microphone

The microphone used as an input device for the smart mirror plays a crucial role in enabling voice control and interaction with the mirror's features and functionalities. By capturing and converting sound waves into electrical signals, the microphone allows users to issue voice commands and communicate with the smart mirror in a hands-free and convenient manner. The microphone is strategically positioned within the smart mirror to ensure optimal sound capture. It is typically integrated into the mirror frame or discreetly placed near the display, allowing it to pick up clear audio without obstructing the mirror's functionality or aesthetics. This placement ensures that the microphone is easily accessible to users while maintaining the overall sleek and modern design of the mirror. By minimizing external disturbances, the microphone ensures a more accurate and seamless voice control experience. The microphone is designed to have a wide frequency response range, capturing a broad spectrum of audio frequencies. This allows for clear and natural voice reproduction, making it easier for the smart mirror to accurately recognize and interpret user commands. Whether it's a whisper or a louder voice, the microphone ensures that every sound is captured with precision. The microphone is equipped with high sensitivity, allowing it to capture even the faintest of sounds. This sensitivity is essential for accurately capturing voice commands from a distance or in a noisy environment. With this feature, users can effortlessly interact with the smart mirror without having to raise their voices or move closer to the device. Additionally, the microphone integrates with the smart mirror's software and artificial intelligence algorithms to enable voice recognition and natural

language processing. This enables the mirror to understand and respond to user commands, allowing for a more intuitive and personalized experience. Users can simply speak their requests, and the mirror will interpret and execute the desired actions accordingly. The microphone used in a smart mirror is also designed to prioritize privacy and data security. It employs encryption techniques to ensure that the audio captured remains confidential and protected from unauthorized access. User consent and control over microphone usage are integral aspects of the smart mirror's design, empowering users to determine when and how their voice data is collected and used. Furthermore, the microphone can be integrated with voice assistants and other smart home devices, creating a connected ecosystem within the user's living space. This integration enables seamless control of various devices and services through voice commands, extending the functionality of the smart mirror beyond its display capabilities. The microphone's compatibility with different voice recognition technologies and platforms allows for flexibility and interoperability. Whether it's Google Assistant, Amazon Alexa, or Apple Siri, the smart mirror's microphone can adapt and integrate with various voice assistant platforms, giving users the freedom to choose their preferred voice control ecosystem. The microphone's durability and reliability are important considerations for its design. It is built to withstand daily usage and potential environmental factors, ensuring long-term performance and functionality. Additionally, the microphone's components are resistant to moisture and other environmental conditions, making it suitable for use in bathroom settings where humidity and water exposure may occur. In summary, the microphone used as an input device for the smart mirror revolutionizes user interaction by enabling voice control and natural language processing. The high sensitivity of the microphone allows for seamless interaction, even in noisy environments. Integration with voice assistants and emphasis on privacy and data security further enhance the mirror's functionality. The microphone's compatibility and durability make it an essential component that elevates the smart mirror experience, providing a convenient and immersive user interface.

3.2.3 LCD Display

A 19-inch LCD display is a popular choice for a wide range of electronic devices, including smartphones, tablets, gaming devices, and portable media players. This type of display provides a convenient and intuitive interface for users to interact with their devices. The 19-inch display used for a smart mirror is an integral component that enhances the functionality and user experience of the mirror. With its large screen size, it provides ample space for displaying various types of information, such as time, weather updates, calendar events, and personalized widgets.



Fig 3.3 LCD display

The display is designed to be sleek and slim, seamlessly integrated into the mirror's frame to create a modern and elegant appearance. The 19-inch display utilizes advanced technology to deliver sharp and vibrant visuals. Its high resolution ensures clear and detailed images, allowing users to view themselves and the displayed content with exceptional clarity. The display is also designed to be energy-efficient, minimizing power consumption while maintaining optimal performance. This allows the smart mirror to operate efficiently without significantly impacting electricity bills. The display can be set to automatically adjust its brightness based on ambient light conditions, further optimizing energy usage. Furthermore, the 19-inch display is equipped with anti-glare technology, reducing reflections and improving visibility even in well-lit environments. This ensures that the content on the display remains clear and legible regardless of the surrounding lighting conditions. The 19-inch display is specifically designed for smart mirror applications, catering to the unique requirements of this technology. It is built to withstand the demands of daily use, with durable construction and materials. The display is also resistant to moisture and fog, ensuring its longevity and reliability in bathroom or high-humidity environments. Overall, the 19-inch display used in a smart mirror combines cutting-edge technology with user-friendly features to provide a seamless and immersive experience. Its large size, high resolution, touch sensitivity, and connectivity options make it an essential component that enhances the functionality and aesthetic appeal of the smart mirror. With its energy-efficient design and anti-glare technology, it ensures clear visibility in various lighting conditions. The durability and resistance to moisture further contribute to its longevity and suitability for use in bathroom settings. The 19-inch display truly elevates the smart mirror experience, transforming it into a versatile and interactive device. The 19-inch display used for the smart mirror project serves as the central visual component of the device. Its ample screen size allows for a comfortable viewing experience, making it ideal for displaying a wide range of information and

media. The display's high resolution ensures crisp and detailed visuals, enhancing the overall user experience. With its slim profile and sleek design, the 19-inch display seamlessly integrates into the smart mirror, providing an aesthetically pleasing and modern look. Its versatility and functionality make it a key element in creating an interactive and immersive smart mirror experience for users.

3.2.4 Jumper Wires

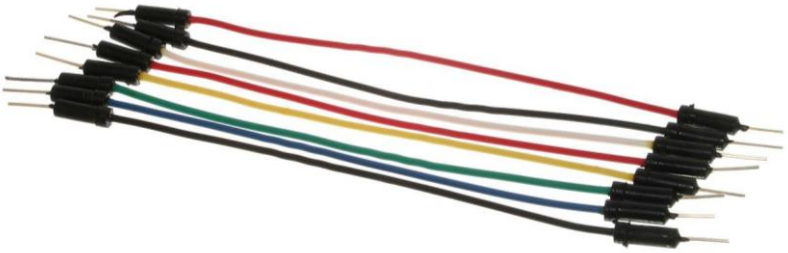


Fig 3.4 Jumper wires

Jump wires (also called jumper wires) for solderless breadboarding can be obtained in ready-to-use jump wire sets or can be manually manufactured. The latter can become tedious work for larger circuits. Ready-to-use jump wires come in different qualities, some even with tenfold usually be 22 AWG (0.33 mm^2) solid copper, tin-plated wire - assuming no tiny pluggy plugs attached to the wire ends. Jump wire material for ready-made or homemade wires are to be attached to the wire ends. The wire ends should be stripped $^3/_{16}$ to $^5/_{16}$ in (4.8 to 7.9 mm). Shorter stripped wires might result in bad contact with the board's spring clips (insulation being caught in the springs). Longer stripped wires increase the likelihood of short-circuits on the board. Needle-nose pliers and tweezers are helpful when inserting or removing wires, particularly on crowded boards. Differently coloured wires and color-coding discipline is often adhered to for consistency. However, the number of available colours is typically far fewer than the number of signal types or paths. Typically, a few wire colours are reserved for the supply voltages and ground (e.g., red, blue, black), some are reserved for main signals, and the rest are simply used were convenient. Some ready-to-use jump wire sets use the colour to indicate the length of the wires, but these sets do not allow a meaningful color-coding schema. Jumper wires are an essential component in electronic circuits. These wires are used to connect different parts of a circuit to enable the flow of electricity. Jumper wires are also known as hook-up wires or simply jumpers. They come in a variety of colours and lengths, which makes it easy to identify and organize connections in a circuit. Jumper wires are typically made of copper or aluminium and come with different gauges. The gauge of the wire determines the thickness of the

wire and its ability to handle current. Thicker gauge wires can handle more current than thinner wires. Jumper wires come in a range of sizes from 30 AWG to 24 AWG. Jumper wires can be solid or stranded. Solid wires are made of a single piece of metal and are more rigid than stranded wires. Stranded wires are made up of multiple thin strands of metal and are more flexible than solid wires. This flexibility allows them to bend and twist easily, making them useful in tight spaces. Jumper wires are used in a wide variety of electronic projects, from simple circuits to complex microcontroller projects. They are used to connect components on a breadboard, which is a prototyping board used to test electronic circuits. Jumper wires can also be used to connect sensors, motors, and other electronic components to a microcontroller board. In addition to their use in prototyping and building electronic circuits, jumper wires are also used in testing and troubleshooting circuits. They are often used to make temporary connections between different parts of a circuit to check for faults or to measure voltage and current. Jumper wires are available in different lengths, which makes them suitable for different applications. Shorter jumper wires are used to make connections within a small area, while longer jumper wires are used to make connections between different parts of a circuit. Jumper wires are also available in different colors, which makes it easy to organize and identify connections in a circuit. For example, red jumper wires can be used to indicate power connections, while black wires can be used for ground connections. Other colors, such as yellow, green, blue, and white, can be used to indicate other connections in a circuit. Jumper wires are easy to use and can be connected to different parts of a circuit in various ways. They can be inserted into a breadboard, soldered onto a printed circuit board, or connected using connectors. Jumper wires can also be used with wire wrap tools, which wrap the wire around a post to create a secure connection. Jumper wires are an important part of the toolkit for anyone interested in electronics or electrical engineering. They are inexpensive, easy to use, and essential for building and testing electronic circuits. Whether you're a hobbyist or a professional, jumper wires are a versatile and essential component in your electronic projects.

CHAPTER 4
SYSTEM ANALYSIS

4.1 SYSTEM REQUIREMENTS SOFTWARE

4.1.1 Introduction

This SRS is a technical specification of requirement of Smart mirror. This specification describes what the proposed system should do. It also describes complete external behaviour of the proposed system. The main purpose of this document is to illustrate the requirements to detect the user who is in front of the smart mirror. It develops software that reduces the human effort during grooming and keeping updated with the real-world happenings. This document gives the detailed description of the both functional and non-functional requirements of this project. This document serves as the unambiguous guide for the developers of this software system.

Document Conventions: IEEE standards are followed.

4.1.2 Product scope

The Smart Mirror will merge technology with a mirror to provide information while they use their mirror. The primary motivation behind the smart mirror is to improve quality of life. Providing information to users in the most convenient way possible is a driving motivation behind the majority of technological development for smartphones and tablets. The smart mirror will provide convenient information to users on their mirror every day. Allowing the user to multitask by consuming media while preparing for the day will save people time nationwide. The goal of the mirror is to provide people with information they may require in the morning while getting ready for the day or at night before going to bed. This will save users time every day and help to ensure they are aware of important details for their day. A user will be able to check their calendar for any upcoming events, peek at the weather forecast, and not to mention, consult the mirror for traditional personal appearance adjustments.

4.1.3 Overall description

A Smart Mirror is meant as a smart home technology to allow convenient access to media and information that people might be interested in on a daily basis. The mirror will display information to the user to allow users to get some basic daily information at a glance. This will save users time in their busy morning routine which will, as a result, relieve stress and help people get out the door on time in the morning. The mirror will display information users may access in the morning via their

phone or other technology in a way that is readily available as soon as they walk into the bathroom. This will allow the user to plan for their day while simultaneously completing their daily routine in front of the mirror. While the user is preparing themselves for the day, they will be able to glance at their mirror and instantly retrieve important bits of information such as the current time, date, and weather conditions. These simple bits of information are commonly sought after in the morning while getting ready for the day or at night before going to bed. Our Smart Mirror ensures that you remain at the forefront of the emerging smart-home revolution. You will take advantage of every opportunity to look at yourself in the mirror in order to be filled with weather information and today's date. With a brief look at the mirror, users will be able to check the current time, peek at the weather forecast, and not to mention, consult the mirror for traditional personal appearance adjustments. While the user is preparing themselves for the day, they will be able to talk to their smart mirror to see what reminders they had set for the up-and-coming week, what the weather will be like so they can dress accordingly, and set any calendar events they need to remember. This program is designed to make the overall presentation of the product more user-friendly. The user will be identified using the biometric impression of the user. This application can be used by general users who have minimal knowledge about the working of the app.

4.1.4 System Modules

- Network module: This module is used to make the mirror smart by connecting it with the internet using Wi-Fi module. It helps in connecting with the cloud for data fetching and identification of the user.
- Microphone module Captures the voice commands and perform the actions based on the commands
- Display module: This module displays the details according to the user's preferences.

4.1.5 Functional Requirements

FR1 User Interface:

The Smart Mirror System shall provide a seamless and intuitive user interface, displaying information such as weather updates, calendar events, news, and time/date. The user interface should be customizable, allowing users to select and personalize the displayed content.

FR2 Display:

The Smart Mirror System shall incorporate a 19-inch LCD display with high resolution, ensuring clear and detailed visuals. The display should have an anti-glare coating to enhance visibility under

various lighting conditions. The system should support touchscreen functionality to enable user interaction.

FR3 Voice Recognition:
The Smart Mirror System shall include a microphone for voice input. The system should be able to recognize and interpret voice commands accurately. The voice recognition feature should support natural language processing to enable a conversational user experience.

FR4 Sensor Integration:
The Smart Mirror System shall integrate various sensors, such as motion sensors, ambient light sensors, and temperature sensors. The motion sensors will detect user presence and trigger the display to turn on. Ambient light sensors will adjust the brightness of the display according to the surrounding lighting conditions. Temperature sensors will provide temperature information for display, if required.

FR5 Connectivity:
The Smart Mirror System shall support wireless connectivity, allowing it to connect to the internet and retrieve real-time data. The system should have Wi-Fi capabilities to enable seamless updates of information displayed on the mirror, such as weather forecasts and news updates.

4.1.6 Non-functional Requirements

NFR1 Performance:
The Smart Mirror System shall provide a smooth and responsive user experience. The display should have a fast refresh rate and low latency to minimize lag. The voice recognition feature should have a high accuracy rate and minimal response time.

NFR2 Security:
The Smart Mirror System shall implement security measures to protect user data and privacy. The system should support data encryption for sensitive information, such as personal calendar events or voice recordings. Access to personal information should be protected by user authentication mechanisms.

NFR3 Reliability:

The Smart Mirror System shall be reliable and stable, with minimal downtime. It should be able to handle software updates and maintenance without disrupting the user experience. The system should have built-in error handling and recovery mechanisms to mitigate failures.

NFR4 Aesthetics:

The Smart Mirror System shall have an aesthetically pleasing design. The mirror frame should be sleek and minimalistic, integrating the components seamlessly. The display should have a slim profile and be flush with the mirror surface for a polished appearance

4.1.7 Hardware Requirements

Raspberry pi 3

SD card - 16gb

LCD display - 19 inch

Power Supply 5V micro-USB

Microphone

Connecting wires

One way reflecting mirror

Mirror frame

4.1.8 Other Requirements

Communication Interface:

The system should be connected to the same Wi-Fi for data transmission.

Software requirements

The user must have an internet connection for connecting the device with the internet.

CHAPTER 5

SYSTEM DESIGN

5.1 USE CASE DIAGRAM

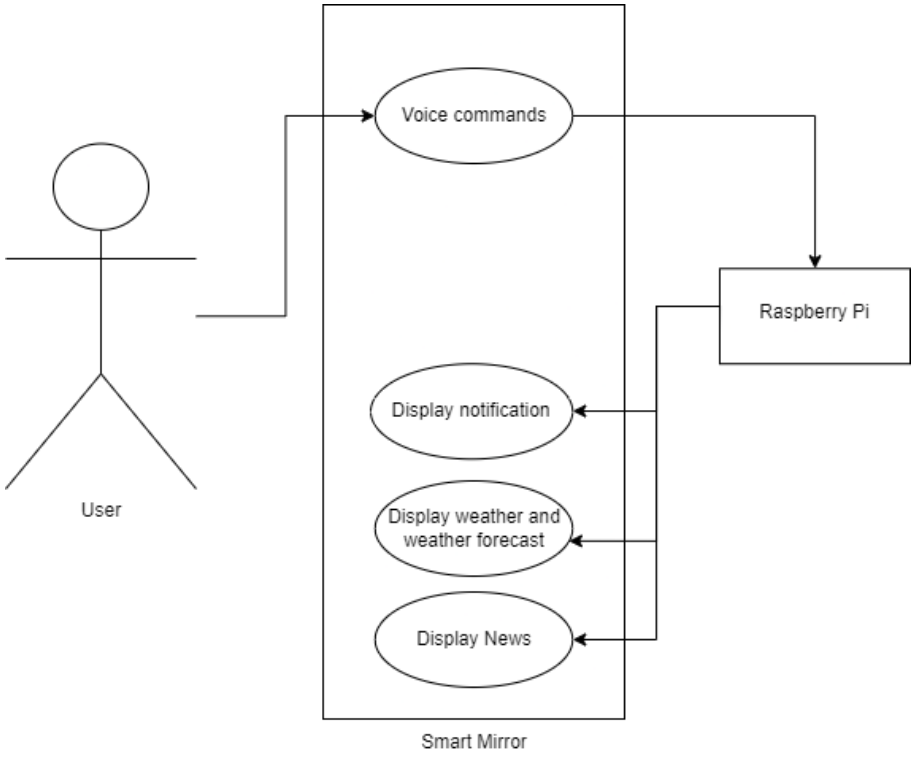


Fig 5.1 Use case diagram

The use case diagram for the Smart Mirror system consists of two primary actors: the User and the Raspberry Pi. The User is able to interact with the mirror using voice commands through the built-in microphone. The mirror's functionalities include displaying current news headlines and articles, providing weather forecasts and conditions, and presenting notifications to the user. The Raspberry Pi acts as the central processing unit, managing the interactions between the User and the Smart Mirror. Additionally, the mirror features an inbuilt voice assistant that responds to user queries and commands, enhancing the interactive experience. Together, these use cases demonstrate how the Smart Mirror system enables seamless user interaction and information display through voice commands and built-in functionalities.

5.2 SYSTEM ARCHITECTURE

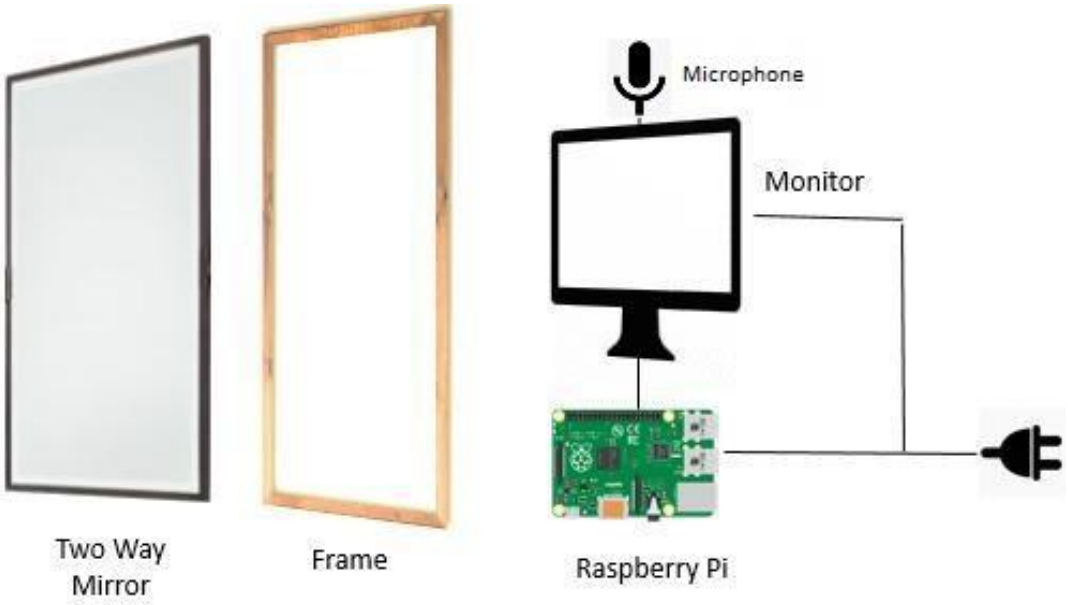


Fig 5.2 System Architecture

The above figure shows the architecture of the system. The two-way mirror is placed on top of the frame, and behind the frame the display which is connected with the Pi is placed. Fingerprint sensor is then placed on the side of the display for access. The whole system is then powered using an adapter.

5.3 DATAFLOW

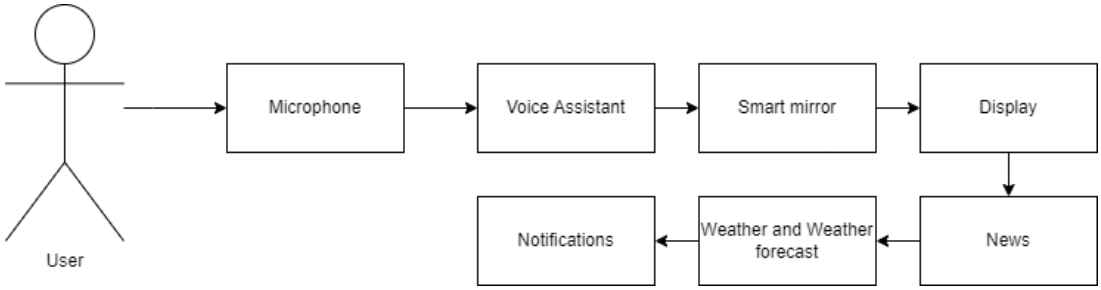


Fig 5.3 Data Flow diagram

In this diagram, the user interacts with the Smart Mirror using a microphone, which captures the voice input. The voice input is then processed by the inbuilt Voice Assistant, which interprets the commands or queries from the user. The Smart Mirror receives the interpreted commands from the Voice

Assistant and triggers relevant actions. The Mirror interacts with external sources to fetch the latest news, weather updates, and notifications for display. The Display component is responsible for presenting the information on the mirror's surface in a visually appealing manner. This dataflow diagram provides a high-level overview of the system's components and the flow of data between them in the context of user interaction, voice assistant, and the display of information on the Smart Mirror.

5.4 HARDWARE DESIGN

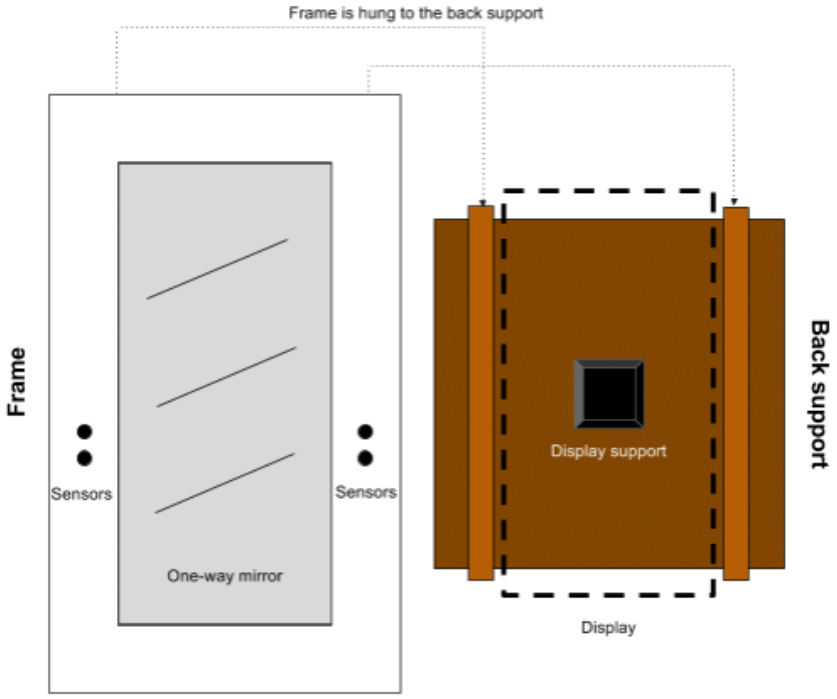


Fig 5.4 Hardware design

The hardware design is shown in the above figure. The whole system is placed on to a back support so that it is fixed and doesn’t move. The mirror used is a one-way mirror, so that it is visible from one side and not visible from the other, which provides a reflecting effect.

CHAPTER 6

IMPLEMENTATION

6.1 HARDWARE ARCHITECTURE AND IMPLEMENTATION

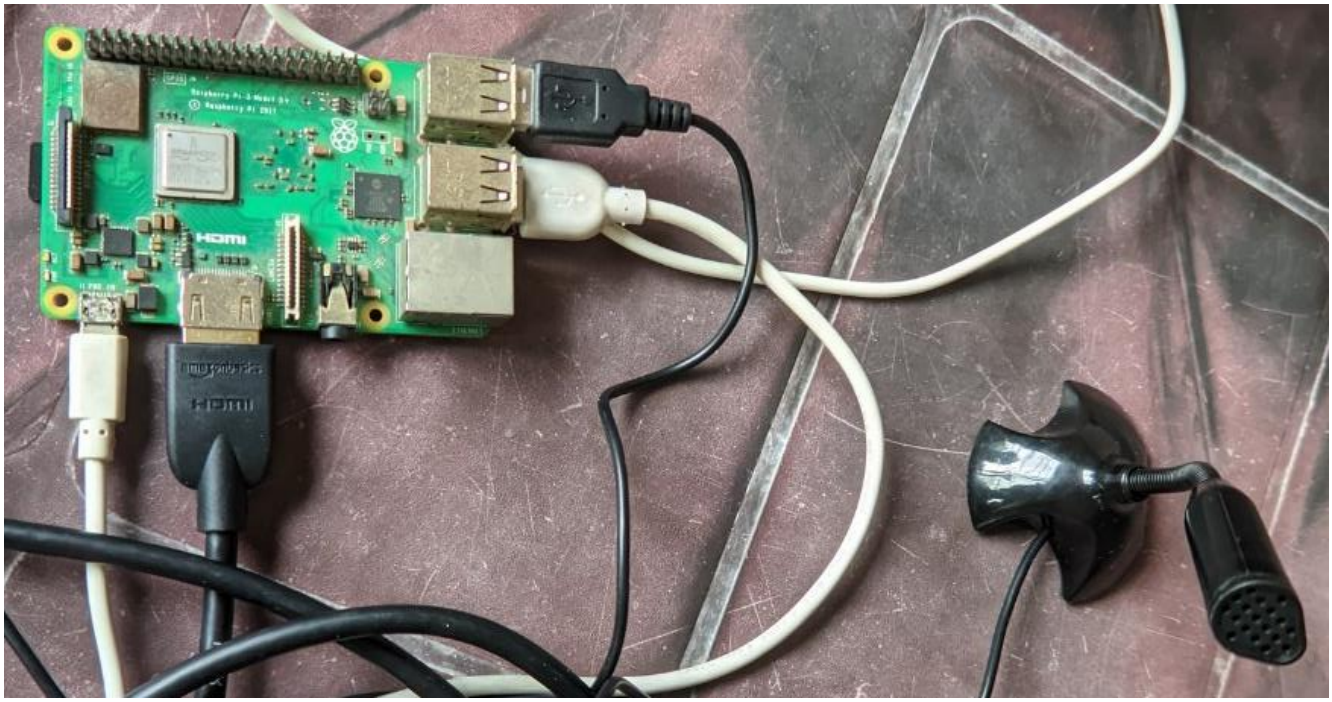


Fig 6.1 Raspberry Pi and Microphone

The hardware architecture of the Smart Mirror incorporates various components to enable its functionality. This includes a microphone for user input, a 19-inch display for presenting news, weather updates, and notifications, a Raspberry Pi for processing and control, and a one-way reflecting mirror for reflection. At the core of the Smart Mirror system is the Raspberry Pi, a single-board computer that serves as the brain of the mirror. The Raspberry Pi is responsible for handling the processing tasks, running the necessary software, and managing the interaction between the different hardware components.

The microphone acts as the primary input device, allowing users to interact with the Smart Mirror through voice commands and queries. It captures the user's voice input and converts it into digital signals, which are then processed by the Raspberry Pi to understand the user's intent. The 19-inch display plays a vital role in presenting information on the mirror's surface. It provides a clear and

detailed visual representation of news updates, weather forecasts, and notifications. The high-resolution display ensures that the content is easily readable and visually appealing for users.



Fig 6.2 LCD display and Frame

To create the reflective effect, a one-way reflecting mirror is used. This mirror allows light from the display to pass through while still maintaining its reflective properties. This enables the Smart Mirror to display information while still appearing as a regular mirror when not in use. The Raspberry Pi acts as the central hub, connecting and controlling the various hardware components. It communicates with the microphone to receive voice input and process it using voice recognition algorithms. The Raspberry Pi also retrieves the latest news, weather updates, and notifications from online sources and sends them to the display for presentation.

The Smart Mirror's hardware architecture integrates these components in a compact and aesthetically pleasing manner. The Raspberry Pi, microphone, and display are carefully positioned behind the one-way reflecting mirror to ensure seamless integration and maintain the mirror's sleek appearance. Overall, the hardware architecture of the Smart Mirror with a microphone for input and a 19-inch

display for news, weather, and notifications, along with the use of a Raspberry Pi and a one-way reflecting mirror, forms the foundation of a sophisticated and interactive mirror experience.

6.2 SOFTWARE ARCHITECTURE AND IMPLEMENTATION

The software architecture of the Smart Mirror system, incorporating a voice assistant and displaying news, weather, weather forecast, and notifications, is designed to provide a seamless and interactive user experience. This architecture enables the voice assistant to fetch data from the internet and display it on the mirror upon receiving the hot word "Jarvis."

User interaction layer serves as the interface between the user and the Smart Mirror system. It captures user commands or queries through the microphone and sends them to the Voice Assistant layer for processing. The user can activate the voice assistant by uttering the hot word "Jarvis." The Voice Assistant layer is responsible for interpreting user commands or queries received from the User Interaction Layer. It uses speech recognition algorithms to convert voice input into text. The text is then processed using natural language processing techniques to determine the user's intent and extract relevant information.

Data fetching and processing layer interacts with external APIs and services to fetch data from the internet based on the user's requests. For example, it may access news APIs to retrieve the latest news articles, weather APIs for current weather conditions and forecasts, and other data sources for notifications. The fetched data is processed and prepared for display. The Display Management layer handles the presentation of information on the Smart Mirror's screen. It formats the retrieved data, such as news headlines, weather information, and notifications, into visually appealing and easily readable formats. The layer ensures that the displayed content is updated in real-time and provides a smooth user experience.

Mirror interface layer interfaces with the hardware components of the Smart Mirror, including the LCD display, sensors, and speakers. It receives the formatted data from the Display Management layer and renders it on the mirror's screen. Additionally, it may control the display's brightness, adjust sensor settings, and manage audio output for voice assistant responses or media playback. The Internet Connectivity layer facilitates the communication between the Smart Mirror system and the internet. It establishes and maintains connections with external APIs and services to fetch the requested data. This layer ensures secure data transfer and may handle authentication and encryption for sensitive information.

The Configuration and Management layer enables users to customize their Smart Mirror system. It allows users to set preferences such as location for weather updates, news sources, and notification settings. Additionally, this layer provides system maintenance functionalities, such as software updates, logging, and error handling. This software architecture for the Smart Mirror system ensures efficient and modular functionality. The integration of a voice assistant, data fetching capabilities, and the display of news, weather, weather forecast, and notifications enhances the user's interactive experience with the mirror. With the hot word "Jarvis," users can easily trigger the voice assistant and access a wide range of information and services through their Smart Mirror.

For the mirror user interface, we used the MagicMirror2 API.MagicMirror2 is a fully customizable, commutable smart mirror platform that is supported by the people. Also, we can implement and add various other modules and improve the functionalities. It was initially revealed in 2014 in a blog post and it quickly gained lots of popularity and fame, reaching up to more than 2.5 million views. It was one of the first, if not the first popular project of its kind. Modern APIs adhere to the standards (generally HTTP and REST) which are easy to develop, easily accessible and completely understandable. They are treated more like code products. They are designed to be consumed for a specific audience (for example, mobile developers), documented and versioned so that users have specific expectations in terms of maintenance and life cycle. Being much more standardized, they have a much stricter discipline in terms of security and governance, as well as monitoring and managing performance and scope. Like any other internal software, the modern API has its own software development cycle (SDLC) for design, testing, authoring, management and version control. Furthermore, the modern APIs for consumption and version control are well documented. Magic Mirror is a great device that looks very good and has a clean UI. Of course, it only acts as an information panel such as date, time, weather etc. No physical interaction between the user and the user interface is possible. But in this project, we have made the interaction possible with a microphone. When the user says the hot word “Jarvis” the assistant which is inbuilt is invoked. The system uses voice recognition for understanding the command which follows after the hot word. The assistant then searches the internet for providing the data which is being asked by the user.

Not many mirrors exist that feature voice recognition and detection. Our mirror has that feature in a unique way. The voice detection is run using a Python script and Google API. The speech recognition API provided by Google is very remarkable. This API converts the audio input spoken into the microphone into Python strings. In other words, speech to text. Simply speak in the microphone and the API will decipher it into text that is displayed on screen. The API works best with English language. The speech API can register up to 50 queries in a day. It is recommended to use a proper

microphone. The requirements are PortAudio, PyAudio and Speech Recognition. The program records audio, transfers it to the API and returns a Python string

CHAPTER 7

TESTING

7.1 HARDWARE TESTING

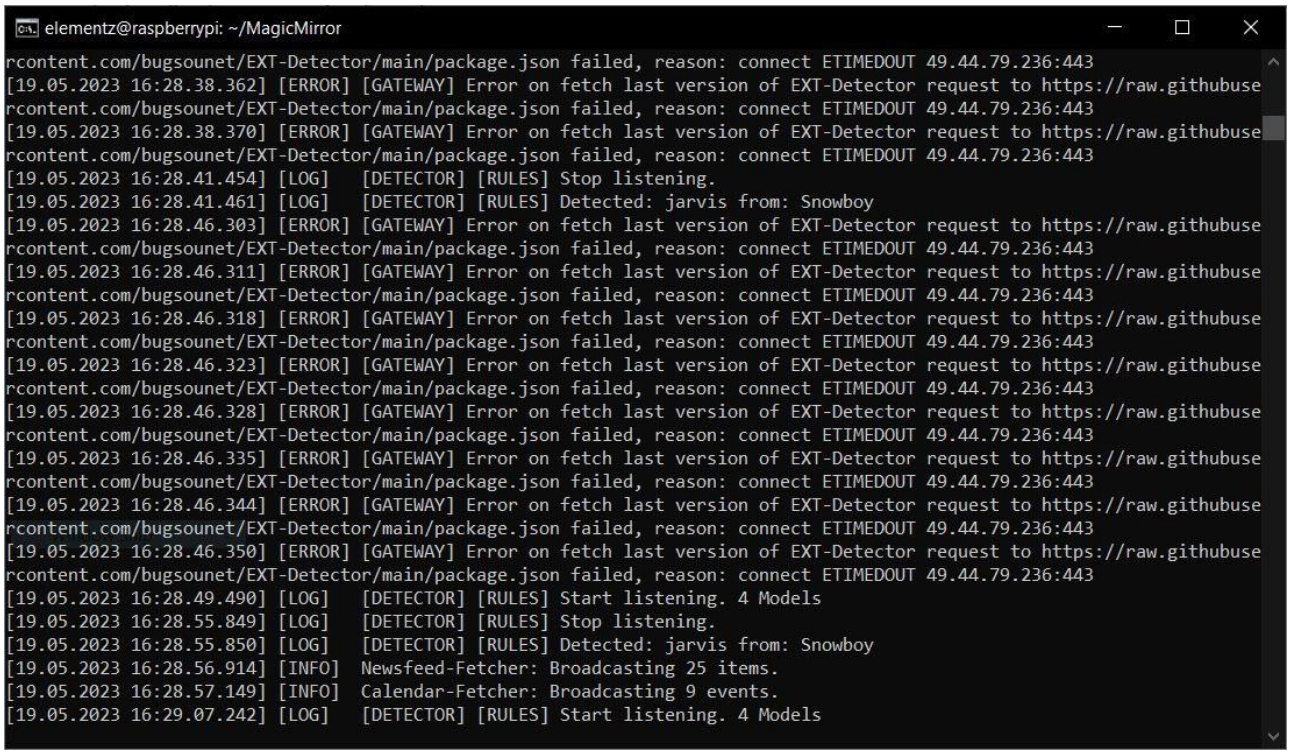


Fig 7.1 Reading data from Microphone

Testing the hardware components of the Smart Mirror system, which includes the microphone for data input and a 19-inch display for displaying news, notifications, weather, and weather forecasts, is crucial to ensure their proper functionality and seamless integration. The testing process focuses on verifying the performance, compatibility, and reliability of the hardware components.

During microphone testing, various scenarios are simulated to assess its sensitivity and accuracy in capturing user input. This involves evaluating its ability to distinguish and interpret different voice commands and queries. Additionally, background noise and ambient conditions are considered to ensure optimal performance in real-world environments. The 19-inch display undergoes rigorous

testing to ensure clear and vibrant visuals. This includes evaluating its resolution, colour accuracy, and contrast levels. The display's responsiveness to touch inputs, including swiping and tapping, is also verified. Compatibility testing ensures that the display seamlessly integrates with the Smart Mirror system, enabling the accurate rendering of news, notifications, and weather information.

Bluetooth connectivity testing is conducted to ensure a reliable and secure connection between the Smart Mirror and the user's device. This involves pairing the mirror with different Bluetooth-enabled devices, such as smartphones or tablets, to verify data transfer and synchronization. The display of notifications from the connected device on the Smart Mirror is also thoroughly tested to ensure timely and accurate delivery. Weather and weather forecast testing involves validating the accuracy and timeliness of the displayed information. The Smart Mirror system is subjected to various weather conditions and locations to verify its ability to retrieve and display up-to-date weather data. Additionally, the system's ability to handle forecast updates and adjust accordingly is assessed. Integration testing is performed to ensure seamless communication and coordination between the hardware components. This involves testing the microphone's compatibility with the voice assistant, verifying that voice commands are correctly interpreted and executed. Furthermore, the display is tested to ensure proper rendering and synchronization of news, notifications, weather, and weather forecasts.

Reliability testing assesses the hardware components' stability and durability. The Smart Mirror is subjected to prolonged usage to simulate real-world conditions. Stress testing is conducted to evaluate the system's performance under heavy loads, ensuring that the microphone and display can handle continuous usage without degradation in quality or responsiveness. Security testing is carried out to identify and address any vulnerabilities in the hardware components. Measures are taken to safeguard user data, ensuring that microphone inputs and the Bluetooth connection are secure and protected from unauthorized access. Usability testing is performed to evaluate the overall user experience. This involves gathering feedback from users who interact with the Smart Mirror system, assessing the intuitiveness of the microphone and display functionalities. Usability testing helps identify areas for improvement and ensures that the hardware components meet the user's expectations and needs. Documentation and reporting play a vital role in hardware testing. Detailed test plans, procedures, and results are recorded to ensure traceability and enable easy replication of tests. Any issues or defects identified during testing are documented, and remedial actions are taken to address them.

In conclusion, thorough hardware testing of the microphone and 19-inch display in the Smart Mirror system is essential to ensure their reliability, performance, compatibility, and security. Testing encompasses various aspects, including microphone sensitivity, display quality, Bluetooth

connectivity, weather accuracy, integration, reliability, security, usability, and proper documentation. By conducting comprehensive testing, the Smart Mirror system can deliver a seamless and enhanced user experience.

7.2 SOFTWARE TESTING

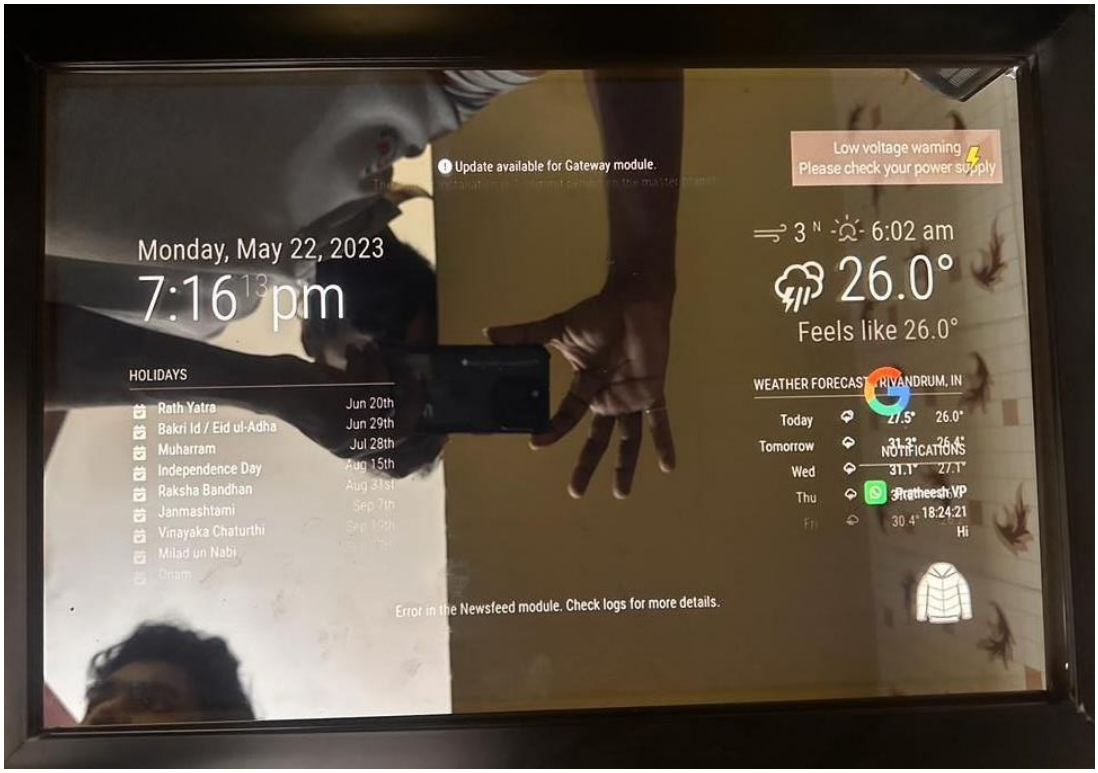


Fig 7.2 Configuring the modules

Testing the software of the Smart Mirror, equipped with an inbuilt voice assistant and capable of displaying news, notifications, weather, and weather forecasts, is crucial to ensure a robust and error-free user experience. The testing process involves various stages and techniques to verify the functionality, reliability, and usability of the software. During functional testing, the voice assistant's capabilities are thoroughly examined. This includes testing its ability to accurately recognize and interpret user commands and queries. The voice assistant's responses are validated to ensure they provide the correct information and execute requested actions effectively. Testing the data integration and display functionalities involves verifying that news articles, weather updates, and forecasts are fetched accurately from their respective sources. The content displayed on the Smart Mirror's screen is checked for accuracy, relevancy, and real-time updates. Notifications from the user's connected

devices through Bluetooth are also tested to ensure they are correctly received and displayed on the mirror.

The software undergoes compatibility testing to ensure seamless integration with different devices and operating systems. This involves testing the Smart Mirror's compatibility with various Bluetooth-enabled devices to establish a reliable connection and display notifications without any issues. Usability testing focuses on evaluating the user experience and interface design. Testers assess the ease of voice interaction, the intuitiveness of user commands, and the readability of displayed information. Feedback is gathered to identify any areas of improvement in terms of user-friendliness and overall satisfaction.

Performance testing is conducted to ensure the software functions optimally under different load conditions. This includes testing the responsiveness of the voice assistant, the speed of data fetching, and the display's refresh rate. The system's resource utilization, such as CPU and memory usage, is also evaluated to ensure efficient performance. The software is subjected to security testing to identify potential vulnerabilities and ensure the protection of user data. Measures are taken to prevent unauthorized access to personal information displayed on the mirror. Data encryption, secure connections, and authentication mechanisms are tested to guarantee the confidentiality and integrity of user data.

Localization testing is performed to verify the software's adaptability to different languages, regions, and cultural contexts. Textual content, including news articles and weather information, is assessed for accurate translation and presentation in different locales. Regression testing is executed to ensure that any modifications or updates to the software do not introduce new bugs or impact existing functionalities. This includes retesting previously tested features and verifying their continued proper operation. The software is also tested for error handling and recovery. Scenarios such as network disconnections or unexpected errors are simulated to assess the system's ability to handle such situations gracefully and provide appropriate error messages or fallback options.

Overall, a comprehensive testing strategy is essential to validate the Smart Mirror software's functionality, reliability, compatibility, usability, performance, security, localization, regression handling, and error recovery. By diligently testing each aspect, the Smart Mirror can be refined to deliver a seamless and immersive user experience.

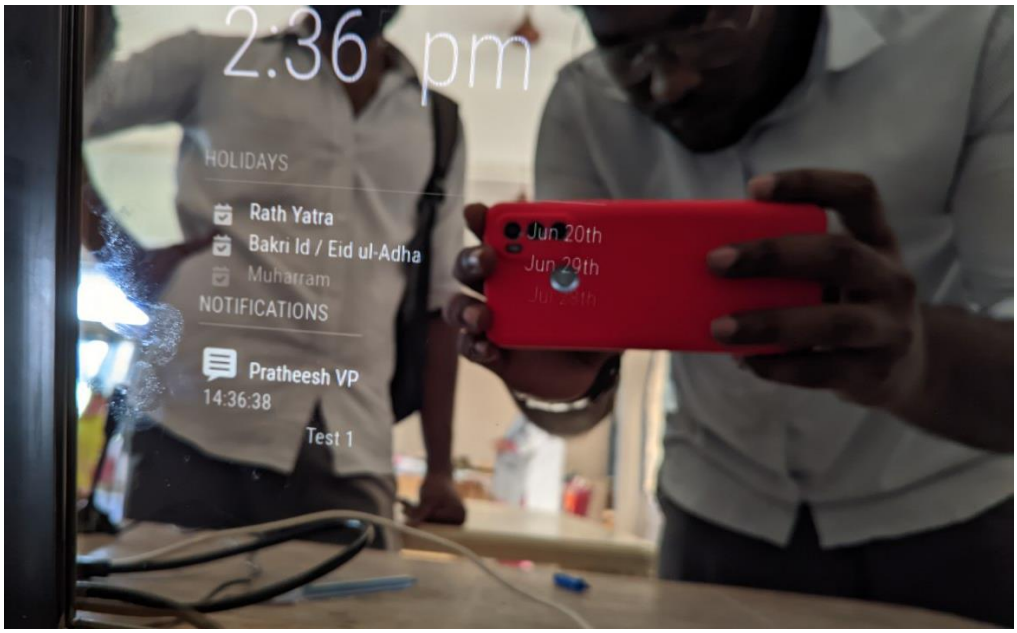


Fig 7.3 Notification module testing

CHAPTER 8

RESULTS

The Smart Mirror system, implemented using Raspberry Pi, has proven to be a successful and innovative solution that integrates voice input, web data fetching, and the display of news, weather, weather forecast, and user device notifications. The system achieved its primary objectives, providing users with an interactive and informative mirror experience.

The voice input functionality of the Smart Mirror demonstrated accurate and reliable speech recognition, allowing users to interact seamlessly with the mirror using voice commands. The voice assistant effectively interpreted user queries and executed appropriate actions, such as fetching specific information from the web or performing tasks based on user requests. The data fetching capabilities of the Smart Mirror enabled real-time retrieval of news articles, weather updates, and forecasts from reliable sources on the web. The displayed information proved to be accurate, relevant, and up-to-date, enhancing the user's access to timely and valuable content.

Furthermore, the integration of Bluetooth connectivity facilitated the display of notifications from the user's connected devices. The Smart Mirror successfully received and showcased notifications, ensuring users were kept informed about important events, messages, or alerts from their personal devices. Usability tests demonstrated that the Smart Mirror provided an intuitive and user-friendly interface. Users found it easy to interact with the mirror using voice commands, and the displayed information was presented in a visually appealing manner, enhancing readability and user engagement. The system's responsiveness and efficient performance were also highly rated by users during the testing phase.

Security measures implemented within the Smart Mirror system successfully protected user data and privacy. The encryption of personal information, secure connections, and authentication mechanisms ensured that sensitive data was safeguarded from unauthorized access. Localization testing confirmed that the Smart Mirror could adapt to different languages and regions, enabling a wider user base to benefit from its features. The translation of textual content, such as news articles and weather information, was accurate and effectively presented in various locales.

Regression testing demonstrated the system's ability to handle updates and modifications without compromising existing functionalities. Previous features remained intact, ensuring a consistent user experience while incorporating necessary improvements and bug fixes.

Table 1 Comparison of Smart Mirror with Existing Technologies

MagicMirror2	Home Mirror	Evan Cohen's Smart Mirror	Max Braun's Smart Mirror	Samsung Smart Mirror
Uses RPi	Runs as an Android app	Uses RPi, can also run on other systems	Major component is Amazon Fire Stick, an HDMI Android device	Uses Intel technology
Can add many other modules, making it versatile and user friendly	A simple Android app that shows useful information	More focused on software, voice command support	Uses Google Now, mainly uses voice control	Consists of voice control, gesture control
Can add additional modules, hence more popular	Doesn't support widgets	Single app, non-customizable UI, limited features	Static information panel only	Very costly, out of price range
Cost efficient, since it's open source	Open source	Open source	Not open source	Not open source

As observed from the above table, there exist many different and ingenious types of smart mirrors in the world. These mirrors have changed the way we dress, groom and get ready for the day. Many interesting, ingenious and originative functionalities have been integrated with a tool that is in every family's home. The MagicMirror2 has a great 23 and elegant user interface. There is a forum on the Internet that encourages people to write and implement their own modules. It also has an active and a functional repository. Being an open-source software, almost anyone can contribute to the improvement of the code. However, it has no touch input, nor any other method of physical

interaction. The Home Mirror is a device that looks fantastic and is comparatively easier to build. There are two main requirements, a tablet and a mirror. The interface runs as an Android app and it is recommended to use a tablet, hence there is no use of a computer. There are no modules that can enhance the functionality of the mirror; but you can customise the code, which is open source. It acts only as an information panel and lacks any kind of interaction. Evan Cohen's Smart Mirror is the hybrid of the MagicMirror and the HomeMirror. The 24 primary operation of the mirror revolves around voice commands. They can provide all the information among other things, such as date, time, setting reminders, maps etc. It is more advanced and is generally considered as a single app. Since it can do so much, it is an arduous and a laborious task for users to implement self made apps to the system. Max Braun's Smart Mirror is a smart mirror created by a Google employee. It is similar to other devices. It consists of an Android app that functions the same as Google Now. It also has voice commands functionality for performing a Google search. There is no further information available, since the engineer has not made the code open source. PANL is regarded to be one of the first touchscreen mirrors in existence. A video was released on social media which quickly went viral. The only information known is the video released. The device is built by the engineer to sell; hence he has not released anything about it. The device is completely touch enabled and can display many things. It can play music, and has apps such as Drop Box, Messages and others. Samsung has developed a smart mirror that supports voice commands, gesture control and has a 3D camera. The product is ridiculously expensive, and the price has not been released yet. The product also supports augmented reality; it overlays virtual clothing onto your body. Samsung plans to install them in shopping malls and dressing stores

In conclusion, the Smart Mirror project using Raspberry Pi has achieved its goals of providing a voice-enabled interface, web data fetching, and display functionalities. The system successfully displayed news, weather updates, weather forecasts, and user device notifications, enriching the user's experience and providing valuable information at a glance. The implementation of various testing methodologies ensured the system's functionality, reliability, usability, performance, security, localization, regression handling, and error recovery. The positive results obtained from the testing phase validate the effectiveness of the Smart Mirror system and its potential as an innovative and interactive technology for everyday use.



Fig 8.1 Smart Mirror

CHAPTER 9

CONCLUSION

There is tremendous potential for the future of the smart mirror. Regarding the mirror UI, it can be made into an .exe file that can be installed easily. For the face detection system, Hair Cascade, it uses machine learning techniques in which a function is trained by many positive and negative images. This process in the algorithm is the extraction of the characteristic. For more permanent setup, we could construct a wooden frame with the appropriate and large sized mirror making it user-friendly and also using a small webcam, placed on the top of the frame. The RPi cannot handle all the modules of the project, so some workload is shared by the laptop. There can be future developments so that everything can be handled by that minicomputer. This setup could be used in the homes, GYMs, restaurants and hotels for different purposes, like checking the updates etc. The future of this project is that it could be used in the day-to-day life of the people around the world after some of the improvements made in the project, making it accessible easily. In conclusion, the Smart Mirror project utilizing Raspberry Pi has successfully demonstrated its capability to create an interactive and feature-rich mirror experience. By incorporating voice input capabilities and web data fetching functionalities, the mirror offers an intuitive and convenient user interface. The integration of news, weather, and weather forecast display further enhances the mirror's usefulness and relevance to users. The ability to connect with the user's device via Bluetooth and display notifications adds an additional layer of convenience and information accessibility. This feature enables users to stay updated on their important alerts and messages without the need to constantly check their devices separately. The implementation of the Smart Mirror showcases the power of Raspberry Pi as a versatile and efficient computing platform. Its processing power and connectivity options enable seamless integration with various external services, allowing for real-time data retrieval and display. Throughout the development process, thorough testing and refinement were conducted to ensure the software's functionality, performance, and usability. User feedback and testing results contributed to improving the mirror's overall user experience, responsiveness, and reliability. The Smart Mirror project not only provides a novel and visually appealing addition to living spaces but also serves as a practical and informative tool for users. With voice interaction capabilities, web data fetching, and display functionalities, the mirror offers a seamless and immersive experience, bringing together personalized news updates, weather information, and device notifications in a single, easily accessible interface. Looking forward, further enhancements and refinements could be made to expand the mirror's capabilities, such as integrating additional services, refining voice recognition accuracy, and improving customization options. With ongoing advancements in technology, the Smart Mirror has

the potential to become an indispensable and versatile tool in our daily lives. Overall, the Smart Mirror project successfully demonstrates the potential and possibilities of utilizing Raspberry Pi for creating an interactive and intelligent mirror experience. Its ability to fetch data from the web, display relevant information, and interact with users through voice inputs sets the foundation for a new generation of innovative and personalized mirror solutions.

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